

Project Title:

*Understanding U.S. Childcare Cost Patterns & UK Pet Cat Movement Through Unsupervised
Analysis*

Section and Team Number:

Section B1, Team 6

Members:

Xinying Wang

Yuanhao Li

Zexuan Wang

Junqi Fang

1. Primary Dataset (Childcare Cost)

1.1 Project Motivation and Preliminary Exploratory Data Analysis (EDA)

Childcare affordability is a significant real-world issue today; it can impact family finances, parents' ability to work full-time, and long-term economic prospects. Childcare costs are quite high in most communities, placing a heavy financial burden, especially on middle-income families who cannot access important subsidies. Using this data can help us to understand differences in childcare costs and affordability across counties in the United States.

Our key stakeholders:

- Parents and families in need of affordable childcare (especially working families)
- Policymakers and government agencies (state/local/federal) that develop childcare subsidies and grant programs
- Employers and human resources departments that may offer childcare benefits to attract and retain employees
- Childcare providers and daycare centers that set prices and determine the scope of their services
- Nonprofit and community organizations that support low-income families and improve accessibility to childcare

Preliminary EDA Summary:

This childcare cost dataset contains 34,567 county-year records from 3,144 counties from 2008–2018 and includes 61 variables. After merging with the county reference table, the final dataset has 64 variables. There is a key limitation in the missingness rate, which is around 31–32% for major childcare cost measures. For example, infant center and home costs are missing, which limits full county-by-year comparisons.

We have five EDA plots to summarize the data. We used yearly `mc_infant` coverage to assess the extent of structured missingness. A national county-level median trend for infant, toddler, and preschool costs. There is a 2018 `mc_infant` histogram to show skew and extreme counties. Also, the 2018 top-15 states by median county `mc_infant` to highlight the regional structure. Last, we have a 2018 scatter of `mc_infant` vs. `mhi_2018`. We find there is a correlation of 0.68 to capture the cost–income relationship and deviations.

1.2 Domain / Business Questions

Question 1: Do U.S. counties fall into a few clear childcare “burden groups” when we look at infant, toddler, and preschool costs together with income and local economic conditions? Where are these groups most common within states and across states? **Decision Value:** This helps policymakers identify county types, for example, there is high-cost/high-income vs. high-cost/low-income, and design more effective support plans. Instead of using a single rule for all counties, we can use states that could create subsidy or childcare supply strategies tailored to different groups.

Question 2: Which counties have childcare costs that are much higher than other counties with similar income levels, and do these counties stay high-cost over time from 2008–2018 or only spike in certain years? **Decision Value:** This could help us to find counties where childcare is unusually unaffordable, not just expensive, because incomes are high. It also helps decide if the county needs long-term support or short-term help during certain years.

Question 3: For counties with enough data, do childcare costs from 2008 to 2018 follow a few common patterns, like steady growth, sharp increases, sudden jumps, or little change? **Decision Value:** This can help leaders detect countries where affordability is getting worse fast and act earlier. It also helps plan resources by focusing on counties that show the riskiest growth patterns.

Question 4: After we account for income and other local factors, are counties still more similar within the same state than across different states in childcare cost levels, burden, and trends? And what does this suggest about state policies versus local market forces? **Decision Value:** This helps us decide whether state-level policies are the main driver of childcare affordability or whether regional markets across states that matters more. The answer can help us guide if solutions should be designed mainly at the state level or through regional coordination.

Who benefits: Those questions would benefit the state and local policymakers, childcare providers, and community organizations by identifying county typologies, persistent high-burden outliers, and shared cost-growth trajectories.

2. Secondary Dataset (Pet Cats UK)

2.1 Project Motivation and Preliminary Exploratory Data Analysis (EDA)

Household cats are one of the most common pets, but when they go outdoors, they may hunt wildlife. This can create a difficult balance between keeping cats happy and safe, respecting owners' wishes, and protecting local animals. Cats often hunt because of instinct and curiosity. However, keeping cats indoors all the time may lead to other problems, such as reduced exercise and boredom. In this project, we are using a UK cat GPS tracking dataset, which could help us better understand cat activity and predation risk. Our goal is to find patterns that can help owners make better choices and reduce harm to wildlife.

Our key stakeholders:

- Pet owners may need practical advice to balance their cat's health and safety.
- Wildlife conservation organizations may seek help to reduce the risk of their cats being preyed upon in dangerous incidents.
- Animal welfare organizations and veterinarians advocate for evidence-based behavioral and care approaches.
- Local policymakers or community groups may consider developing guidelines for outdoor roaming or risk-minimization measures.

Preliminary EDA Summary:

Our dataset consists of two tables: one is an event-level GPS table, which has 18,215 records and 11 variables, and a reference table with 16 attributes for 101 cats. The data is unbalanced because some cats have many more GPS points than others; for example, the range is 12 to 963, and the average of nearly 180. Some records are labeled as outliers by the algorithm with 1.16% and by manual review with 0.77%, and ground_speed has an extremely high value with 277,092. We are thinking it's because of tracking error. Variables such as prey_p_month and hrs_indoors also fall into clear groups, which supports the identification of lifestyle and risk categories.

2.2 Domain / Business Questions

Based on our EDA, we propose is to identify meaningful behavioral patterns of pet cats in the UK. These unsupervised analyses could help us find natural groups, similar behavioral profiles, and unusual cases, rather than predicting a single labeled outcome. Our EDA shows that the cats vary considerably in the number of GPS locations, the length of tracking periods, roaming distances, and maximum speeds recorded.

Question 1: Are cats grouped according to their activity range with bbox_area_proxy, indoor activity time with hrs_indoors, and prey count with prey_p_month? **Decision Value:** If cats are grouped, cat owners and wildlife conservation organizations can intervene, specifically targeting cats that may be harming wildlife. This provides more helpful guidance than applying the same rules to all cat owners.

Question 2: If cats behave similarly, such as in the same locations and speed patterns, are these similarly behaving cats also similar in prey count? **Decision Value:** When similar activity patterns lead to similar prey outcomes, we can categorize cat behavior and provide clear guidance based on behavior type. This helps address the problem if there is limited tracking information for some cats.

Question 3: Are there extremely high-speed records that could be genuine cat activity tracks or tracking errors? What would we find if we removed these records? **Decision Value:** Dealing with extreme values is the right way to avoid erroneous results and incorrect labeling of cats as "high-risk." This makes the conclusions more credible to cat owners and researchers.

Question 4: What variables affect the comparison results? For example, the number of days each cat was tracked and the number of GPS activity counts? Which method is the fairest for comparing cats? **Decision value:** The more data a cat has, the more active it may appear, simply because it has been tracked for a longer period. Fair comparisons reduce bias, leading to better recommendations.

Who benefits: These questions can help pet owners, animal welfare organizations, and wildlife conservation stakeholders better understand cat behavior patterns and predation risks. Overall, these insights can support more targeted and practical recommendations, such as focusing on specific wandering and risk characteristics rather than treating all cats in a one-size-fits-all manner.

3. Appendix

3.1 GitHub Repository Link:

https://github.com/paddington-L831/BA820_TeamA06_project.git

3.2 Supplementary Material

3.2.1 External Sources

Childcare Cost:

NDCP Price Changes by Race and Ethnicity. (2024, September 2). Public.tableau.com.

[https://public.tableau.com/app/profile/women.s.bureau.department.of.labor/viz/NDCPPPriceChangesbyRaceandEthnicity/NDCPPPriceChange](https://public.tableau.com/app/profile/women.s.bureau.department.of.labor/viz/NDCPPriceChangesbyRaceandEthnicity/NDCPPPriceChange)

U.S. Department of Labor. (n.d.). *National Database of Childcare prices*. National

Database of Childcare Prices. <https://www.dol.gov/agencies/wb/topics/featured-childcare>

Pet Cats UK:

Cats, their owners and the Environment - Songbird Survival Research. Cats, their owners and the environment - SongBird Survival Research. (n.d.).

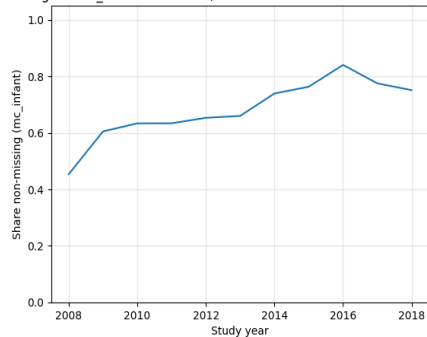
<https://www.songbird-survival.org.uk/categories/cats-and-their-owners>

HL, L., M, B., & M, H. (2025). *What the cat dragged in: Quantifying prey return rates of pet cats (Felis catus) with outdoor access in the UK*. Ecology and evolution.

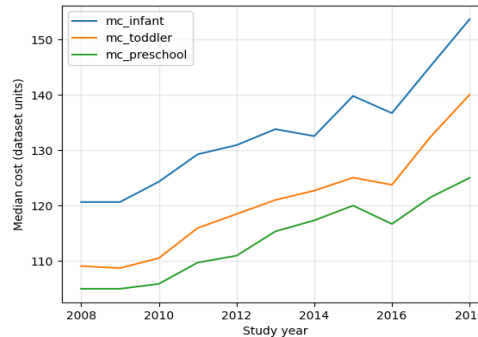
<https://pubmed.ncbi.nlm.nih.gov/40051457/>

3.3.2 Graphs & figures (Childcare Cost)

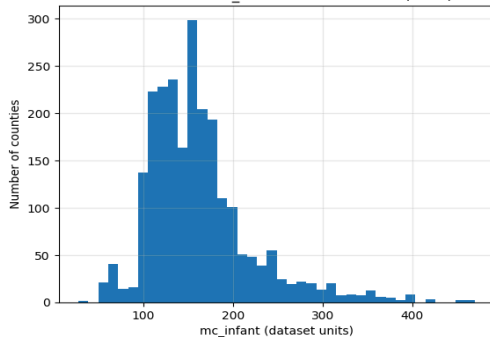
Coverage of mc_infant over time (share of counties with non-missing values)



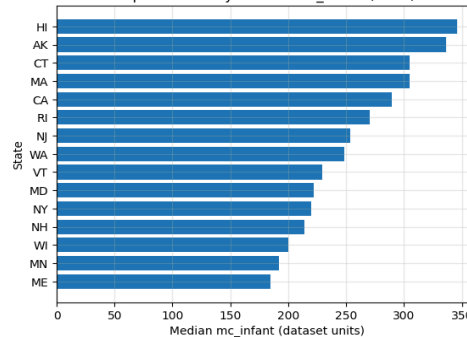
Median childcare cost trend across U.S. counties

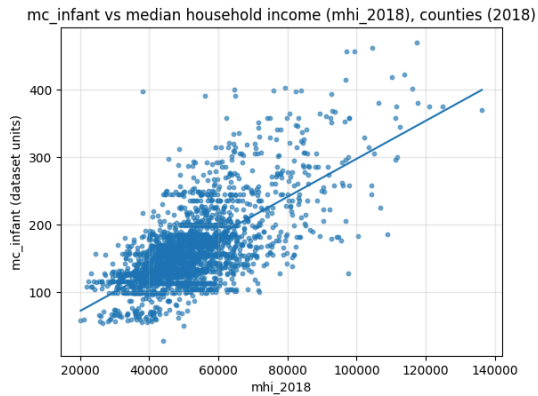


Distribution of mc_infant across counties (2018)

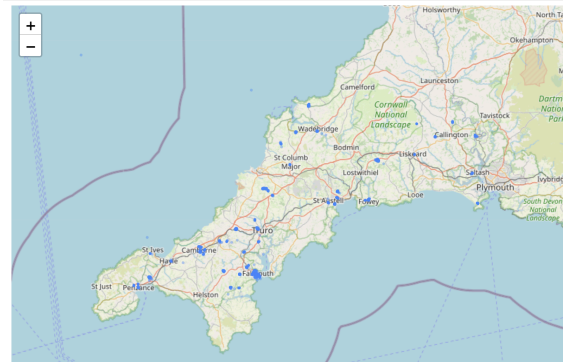
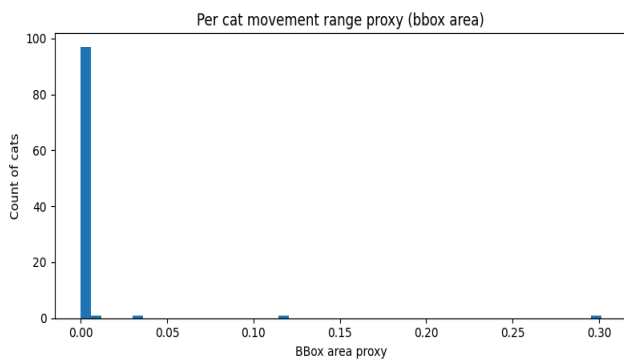
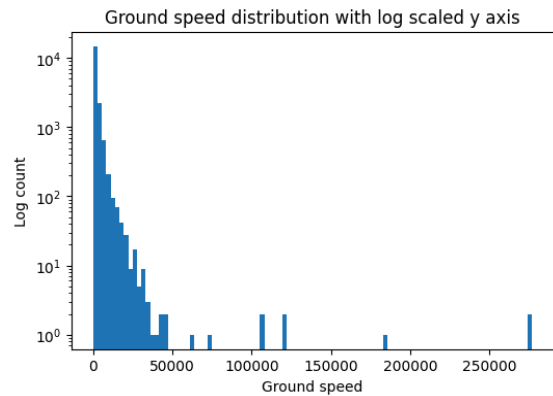
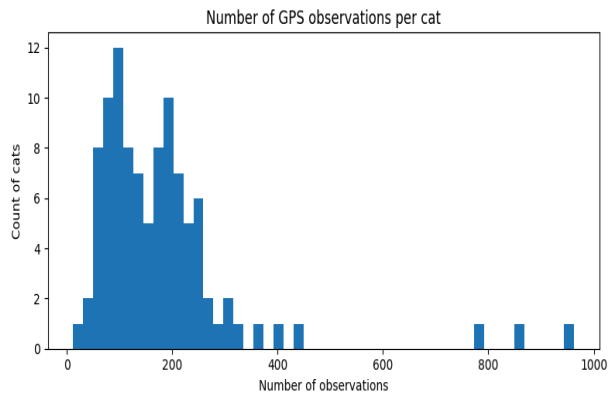


Top 15 states by median mc_infant (2018)





3.2.3 Graphs & figures (Pet Cats UK)



3.3 Process Overview

Our analytics process did follow an advanced exploratory workflow designed to support unsupervised insight discovery:

1. Data Collection and Integration by loading and merging datasets
2. Data Preprocessing, we handled missing values and validated key variables
3. Preliminary Exploratory Analysis, such as using distribution analysis, variability analysis, correlation analysis, and extreme case analysis

4. Insight Generation by identifying patterns, segments, and similarities to domain questions

3.4 Use of Generative AI Tools

We used Generative AI tools throughout the project to brainstorm research questions, redraft proposal outlines to ensure all information was included, refine language, and debug exploratory code. All AI-generated content was reviewed, revised, and integrated by our team.

Here is our ChatGPT link:

<https://chatgpt.com/share/697aaa76-2544-8011-9712-fbdf3de28c77>